분석 시각화

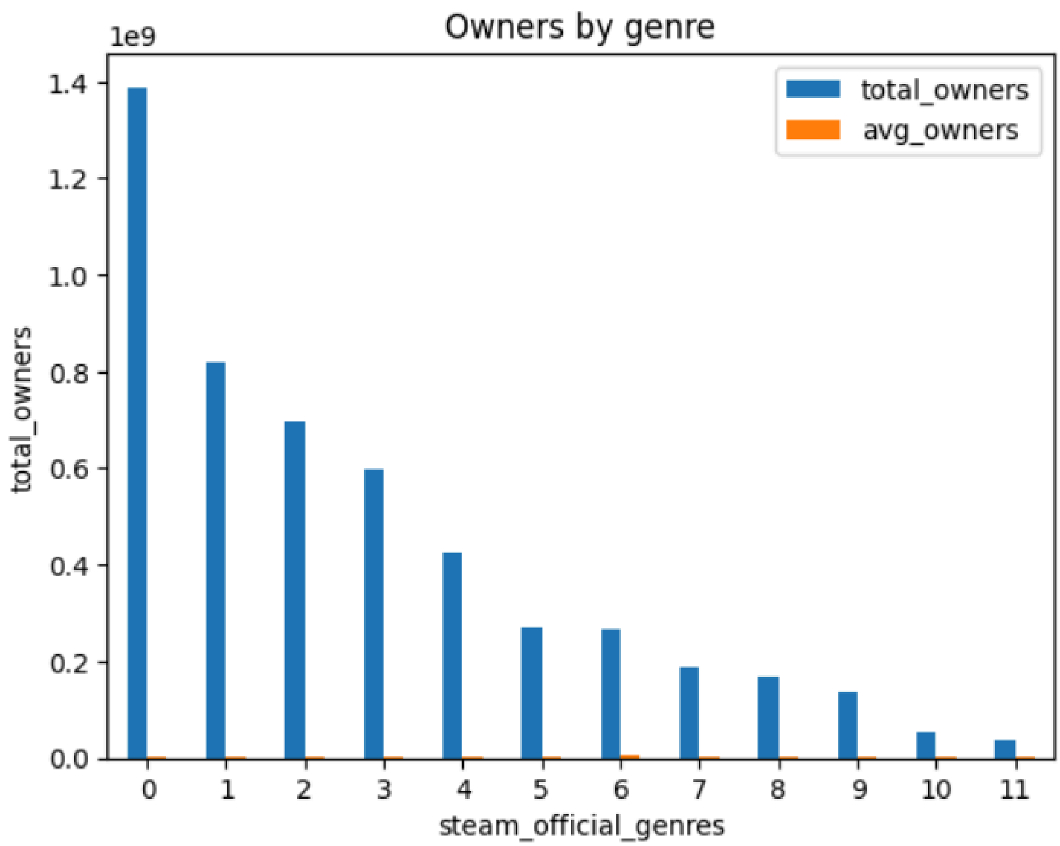
```
df=pd.read_csv("C:/Mydisk/workspaces/SQL_project/genre_user.csv")
df.head()
```

	steam_official_genres	total_owners	avg_owners
0	Action	1388608890	2.760654e+06
1	Adventure	817216260	2.190928e+06
2	Indie	695782950	2.089438e+06
3	RPG	597676890	2.157678e+06
4	Simulation	425611110	1.514630e+06

장르, 총 유저 수, 평균 유저 수만 기록 된 CSV 데이터프레임을 이용.

장르 별 총 유저 수(막대)
x축: 장르, y축: 유저수

```
df.plot.bar(xlabel='steam_official_genres',
            ylabel='total_owners',
            rot=0,
            title='Owners by genre')
plt.show()
```



장르 이름 출력 안됨. 평균 유저수가 함께 출력 됨.

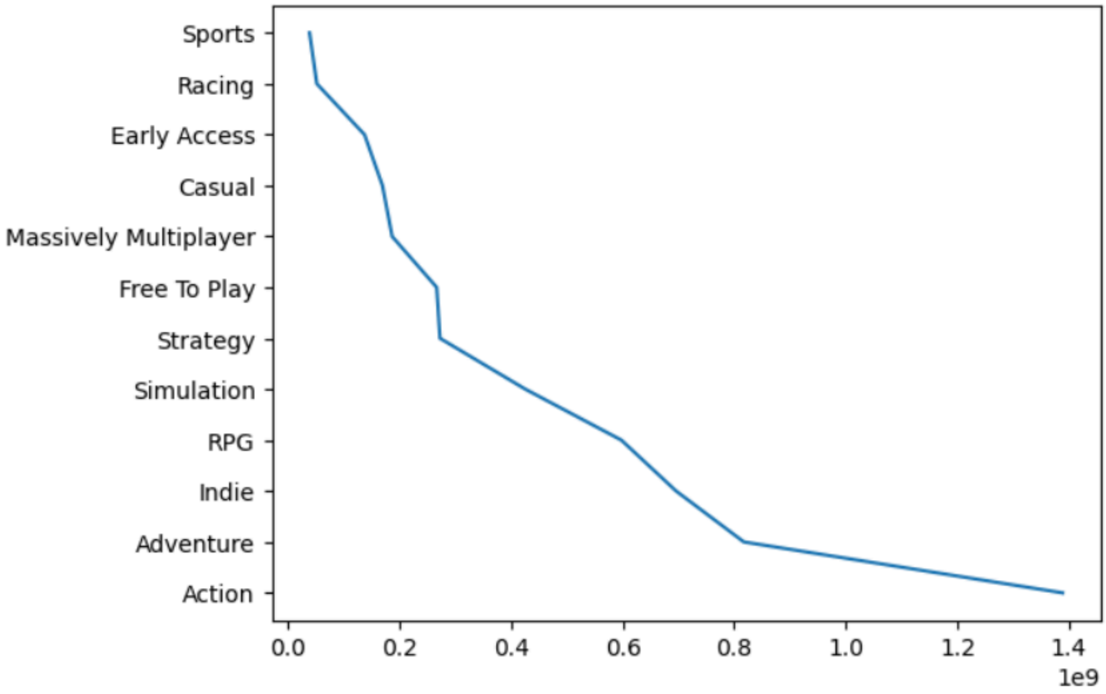
```
column=['steam_official_genres', 'total_owners']
owners_by_genre=df[column].copy()
owners_by_genre
```

	steam_official_genres	total_owners
0	Action	1388608890
1	Adventure	817216260
2	Indie	695782950
3	RPG	597676890
4	Simulation	425611110
5	Strategy	272506170
6	Free To Play	266590830
7	Massively Multiplayer	186719040
8	Casual	169345560
9	Early Access	137200080
10	Racing	51681870
11	Sports	38629380

필요 컬럼만 분리.

```
plt.plot(owners_by_genre['total_owners'], owners_by_genre['steam_official_genres'])
```

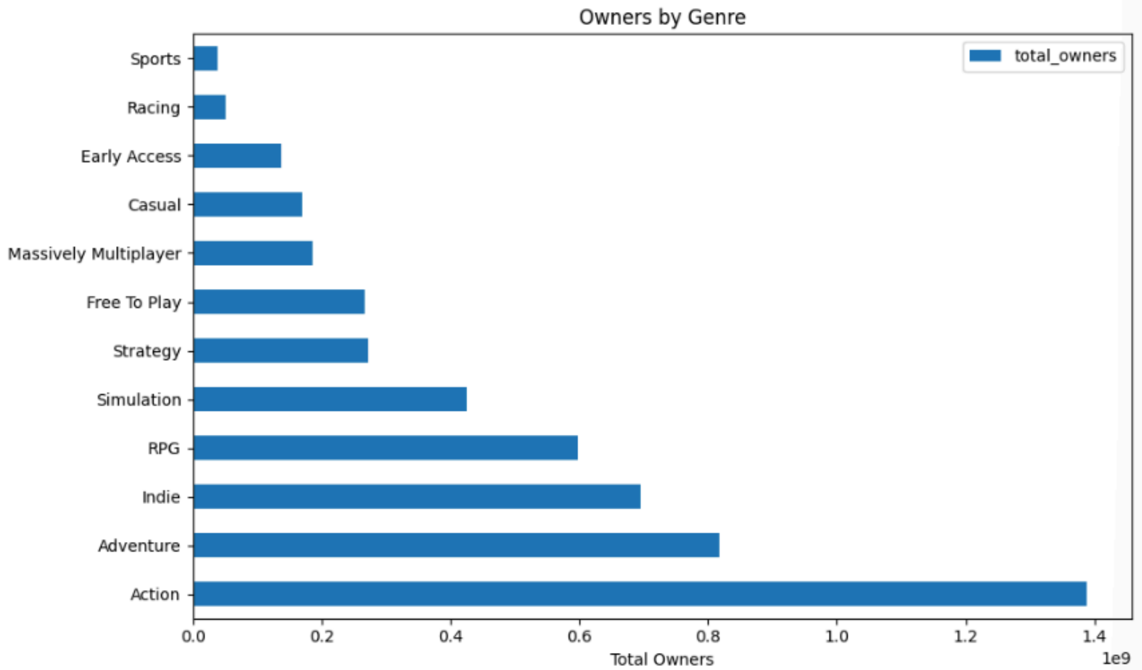
```
[<matplotlib.lines.Line2D at 0x24266267550>]
```



선 그래프 출력. 하지만 원하는 것은 막대그래프.

```
owners_by_genre.set_index('steam_official_genres')[['total_owners']].plot.barh(
    figsize=(10, 6), title='Owners by Genre', xlabel='Total Owners', ylabel=''
)

plt.tight_layout()
plt.show()
```



set_index를 이용하여 y축에 장르 명을 출력. 막대그래프를 채택하여 장르별 유저 분포도를 한 눈에 볼 수 있도록 설정.

장르 별 평균 유저 수 및 긍정평가 비율(산점도)
x축: 유저수, y축: 긍정평가 비율

```
df2=pd.read_csv("C:/Mydisk/workspaces/SQL_project/genre_avg.csv")
df2
```

	steam_official_genres	total_owners	avg_owners	avg_positive
0	Action	1388608890	2.760654e+06	82.3141
1	Adventure	817216260	2.190928e+06	83.5282
2	Indie	695782950	2.089438e+06	87.1201
3	RPG	597676890	2.157678e+06	81.1877
4	Simulation	425611110	1.514630e+06	84.6833
5	Strategy	272506170	1.449501e+06	83.4787
6	Free To Play	266590830	5.126747e+06	69.5962
7	Massively Multiplayer	186719040	2.917485e+06	71.7969
8	Casual	169345560	1.282921e+06	86.6970
9	Early Access	137200080	1.193044e+06	82.8696
10	Racing	51681870	1.292047e+06	81.9250
11	Sports	38629380	9.197471e+05	78.3810

총 유저수, 평균 유저수, 평균 긍정 평가 비율이 포함된 데이터프레임 생성.

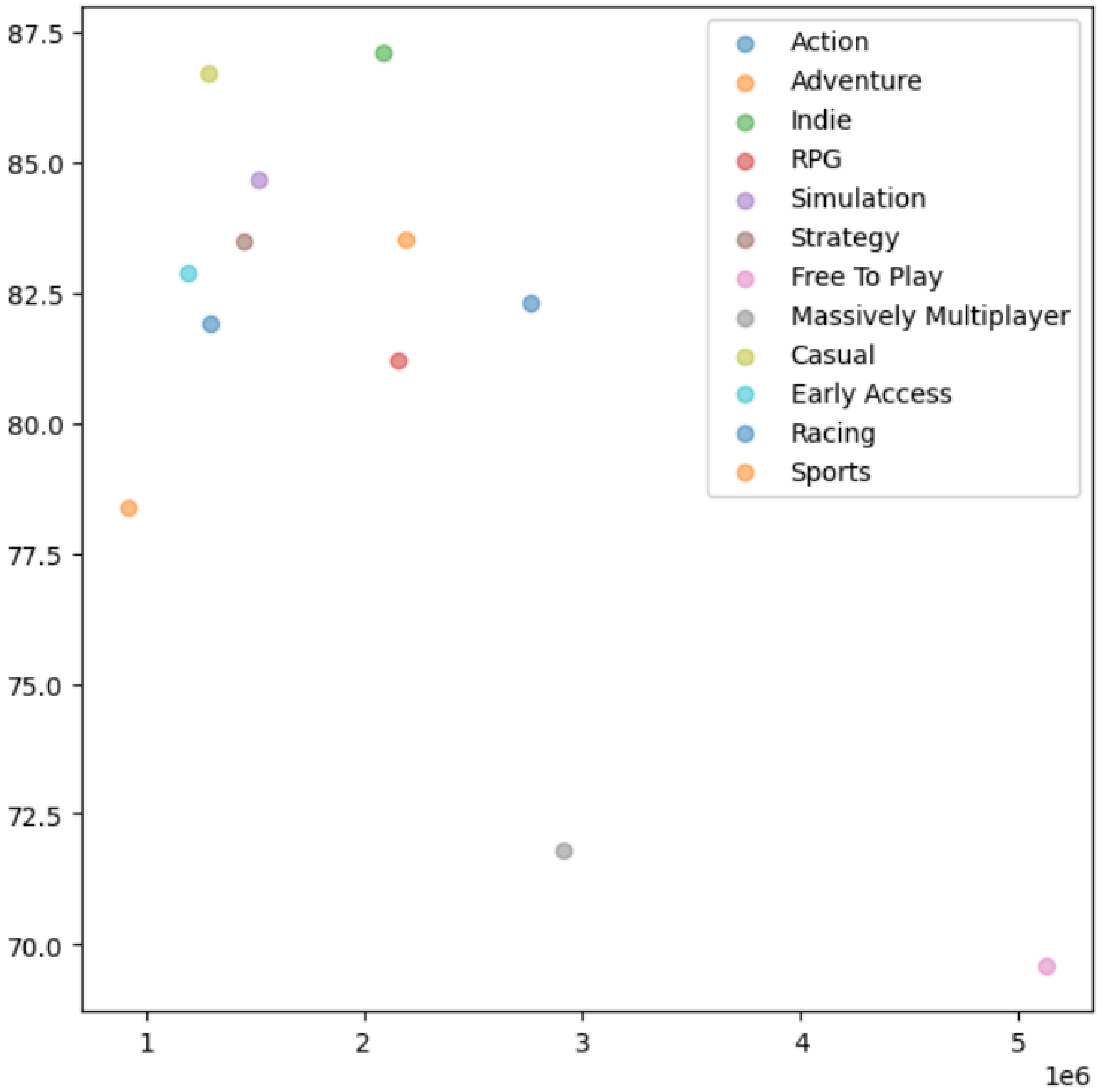
```
column=['steam_official_genres', 'avg_owners', 'avg_positive']
genre_avg=df2[column].copy()
genre_avg
```

	steam_official_genres	avg_owners	avg_positive
0	Action	2.760654e+06	82.3141
1	Adventure	2.190928e+06	83.5282
2	Indie	2.089438e+06	87.1201
3	RPG	2.157678e+06	81.1877
4	Simulation	1.514630e+06	84.6833
5	Strategy	1.449501e+06	83.4787
6	Free To Play	5.126747e+06	69.5962
7	Massively Multiplayer	2.917485e+06	71.7969
8	Casual	1.282921e+06	86.6970
9	Early Access	1.193044e+06	82.8696
10	Racing	1.292047e+06	81.9250
11	Sports	9.197471e+05	78.3810

필요 컬럼 추출.

```python
fig=plt.figure(figsize=(7,7))
ax=fig.add_subplot(111)

for genres in genre_avg['steam_official_genres'].unique():
    sub=genre_avg[genre_avg['steam_official_genres']==genres]
    ax.scatter(x=sub['avg_owners'],
               y=sub['avg_positive'],
               alpha=0.5,
               label=genres)
ax.legend()
plt.show()
```



산점도 출력.

총 유저 수: 액션 장르가 압도적.
긍정적 평가 비율의 경우, 인디 게임이 가장 높은 비율을 보임.
무료게임의 경우 평균 유저수는 가장 많으나, 평균 평가는 가장 낮은 수치를 보임. 무료라는 이유로 시작했다가 실망하는 경우가 많은 것으로 추정.